

Behind The Scenes

A lot of us have heard of OpenGL and DirectX—especially those of us into 3D gaming. But what do they do?

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Most of us have our favourites when it comes to games. A few of us are crazy enough to spend big bucks on oodles of RAM and a top-of-the-line graphics card, just to be able to play 3D games—others are content with *Solitaire*! If *Solitaire* is your thing, we should warn you that this article could induce sleep; those of you who swear by games such as *FarCry*, *Doom 3*, *Half-Life 2* and *Call Of Duty 2* will find the following informative.

Think about your favourite 3D game. Now think about all the complexities that go into

programming it. First and foremost, you have textures and shadows and lights and water effects and... the list is endless. Games today come up with breathtaking scenery, and this is expected of any big title worth its salt. Add to this mix the even more complex physical properties that each object needs to exhibit while you're playing the game, and you have a maddening amount of mathematics, coding and rendering that the graphics chip and CPU have to do in parallel.

Today, only very small bits of scenery are two-dimensional images, if at all. Most of the awe-inspiring fantasy worlds you see in games are made up and rendered on the fly by your graphics subsystem.

